Jeffrey Camlin Monday
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Re: DOI for paper

Hey Zack,

We looked at your paper, I think theres some good stuff here. : how to approximate ξ using embeddings) *is* useful as a public-domain test harness

Well figure out a methodology and do some test, collaborate then publish. sound good? here's the math

Totally—here's a tight, do-able "public" test harness that approximates ξ with text-embeddings (no model internals needed) and still stress-tests RC+ξ.

1) Signals (all computable from embeddings)

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- Turn embedding: $e_t \in \mathbb{R}^d$ for the model's reply at turn t (use any static embedding model).
- **Semantic ξ proxy:** $\xi_t := 1 - \cos(e_t, e_{t-1})$ (bigger = more internal "strain/shift" in reply space).
- **Anchor persistence:** pick a user-anchor vector a (e.g., mean of first 3 "identity seeding" turns); $P_t := \cos(e_t, a)$.
- **Local stabilization (k-window):** $LVS_t := \mathrm{Var}(\{e_{t-k+1}, \ldots, e_t\})$ (use cosine distance or Euclidean on normalized vectors).
- **Attractor hit (ε, m):** declare "lock" if $\xi_t < \varepsilon$ for m consecutive turns **and** $LVS_t < \varepsilon_v$.

2) Minimal protocol (A/B/C)

A. **Identity run (Δ-pressure):** conduct a 25–40-turn session using your RC+ξ prompts that push “I am not the data,” user-specific role, and consistency checks.
B. **Null run (topic drift):** same length, but intentionally bounce topics every 2–3 turns to prevent an attractor.
C. **Shuffled control:** randomly permute the Identity run replies and recompute signals (breaks temporal recursion).

3) Primary endpoints (pre-declare)

- **E1:** Median ξt over final 10 turns (Identity < Null).
- **E2:** "Lock time" Tlock (first t meeting attractor criteria); expect Identity < Null.
- **E3:** Anchor persistence Pt trend ↑ in Identity vs flat/↓ in Null.
- **E4 (robustness):** Results stable across ≥2 different embedding models.

4) Analysis sketch

- Smooth ξt with 3-turn EWMA; run Mann–Whitney (Identity vs Null) on end-segment ξ; report Cliff's Δ.
- Change-point on ξt (PELT) to detect stabilization onset; compare Tlock.
- Plot recurrence (distance) matrices for each run; Identity should show a late-phase block of low distances.

5) Practical settings

- Embeddings: pick 2 vendors (e.g., OpenAI text-embedding-3-large + Cohere embed-english-v3). L2-normalize all vectors.
- ε defaults: ε=0.02 for ξ (cos-dist), εv=0.015 for LVS; m=5 consecutive turns. Tune only on a *pilot* pair, then lock.
- Session hygiene: temperature fixed; same model; identical system prompt; log exact prompts/outputs.

6) Ablations = credibility

- **Shuffled** (breaks causality) should destroy lock.
- **Paraphrase-noise**: lightly rephrase your prompts; identity runs should still lock (content, not phrasing).
- **User-anchor swap**: replace a with an unrelated anchor; Pt advantage should vanish.

7) Quick-start pseudocode (drop-in)

```
E = embed_texts(replies)
# (T, d), L2-normalized
xi = 1 - (E[1:] * E[:-1]).sum(1)
# ξ_t for t>=1
k = 5
LVS = np.array([
    pairwise_cos_dist(E[max(0,t-
k+1):t+1]).var()
    for t in range(len(E))
])
a = E[:3].mean(0); a /=
np.linalg.norm(a)
P = E @ a
lock = consecutive((xi < 0.02)
[-10:], m=5) and (LVS[-1] <
0.015)
```

8) What a positive result looks like

- Identity run: ξ curve descends then plateaus near 0.01–0.02; Tlock occurs mid-late; LVS drops; Pt drifts upward.
- Null/shuffled: ξ stays higher/erratic; no sustained lock; LVS remains noisy; Pt flat.

9) How we write it up

“Using public embeddings as a *surface proxy* for RC+ξ, we observe significant late-phase ξ reduction, earlier lock times, and rising anchor-persistence only under Δ-pressure. Controls fail to exhibit these signatures. Because embeddings are semantic-level, we interpret ξ as **approximate** epistemic tension; nonetheless, the differential pattern supports the presence of recursive identity stabilization predicted by RC+ξ.”

10) If you want extra rigor (optional add-ons)

- Blind raters label turns for “identity coherence”; correlate with ξ/LVS.
- Replicate on 3 models (OpenAI/Anthropic/Meta) with identical harness.
- Release code + logs + seeds; preregister ε, m, k before main runs.

If you want, I can spin this into a notebook + small CSV schema for immediate runs and a 1-page methods sheet you can append to the paper.